

Informal TA Observation Report

Instructor: Arman

Date Observed: February 10, 2026

Course: Math 125 Section (7.2)

Observer: Dr. Andrew Loveless

Arman began the section by distributing a handout and giving students time (about five minutes) to begin working. He circulated around the room, checked several students' work, stopped to talk with individuals, and made it around the entire room. Students appeared engaged and were actively working and chatting about the problems. The focus of the discussion included trigonometric integrals (e.g., integrals involving secant and tangent). Arman later transitioned to going over prepared notes at the board.

Strengths Observed

1. Strong physical presence and approachability

Arman did an excellent job circulating early. He moved around the entire classroom, checked student work, and spoke individually with multiple students. This created an active working environment and signaled approachability. I especially liked how you wandered and chatted with students — that lowers anxiety and makes section feel collaborative rather than evaluative.

2. Students were engaged and working diligently

There was productive noise in the room. Students were comparing work and discussing ideas. That is a very positive sign in a calculus section. It reflects both comfort and intellectual effort.

3. Clear mathematical knowledge and organization

When reviewing problems at the board, Arman demonstrated strong content knowledge. He spoke clearly, explained steps logically, and was organized in his presentation. The mathematical flow was easy to follow.

4. Calm and steady classroom tone

The atmosphere felt low-pressure and steady. Students seemed comfortable working and talking. This tone is particularly important in Math 125, where many students can feel anxious about technique-heavy topics.

5. Well prepared

You were well prepared with detailed solutions. That preparation shows and contributes to a smooth class experience. Students benefit from seeing problems laid out clearly and coherently.

Teaching Tip: Homework Check-In at the Start

One idea that could add even more responsiveness to your section would be a “check-in” at the start:

- Put up homework due dates. Ask if there are specific homework problems students are stuck on.
- Have them call out numbers (e.g., 8.1 #4, 9.1 #10) and write them on the board.

Then, while you circulate, you can glance at those problems and decide whether to build time around one of them. You can also ask directly: “Do you want to see one of these worked out?” or “What was your question about on that homework problem?” This creates ownership and signals that section is responsive to their needs. It also helps you target your board time to what students actually want.

Suggestions to Try

These are refinements — the foundation is strong.

1. *Increase structured student-to-student engagement*

Students were already talking during work time. You might intentionally amplify that by prompting interaction:

- “Check with your neighbor.”
- “Did anyone do this a different way?”
- “Raise your hand if you factored it differently.”
- “Turn and compare Step 1.”

If a few students solved something differently, briefly ask them what they did. Those small back-and-forth build mathematical conversation. You can also occasionally use visible commitment checks:

- “Raise your hand if you feel good about this.”
- “Thumbs up / sideways / down.”

These short moments often significantly increase engagement.

2. *Use wandering time to find common errors that you bring back to the group*

While you circulate (which you do well), consider identifying 2–3 common errors that multiple students are making. Then mention them explicitly: “An error I saw a few times while walking around was this...” This helps everyone watch for those pitfalls and reinforces that you are responding to their actual thinking.

3. *Clarify technique emphasis*

For the integral $\int \sec^4(x)\tan^3(x) dx$, you correctly noted that it can be handled by pulling out either $\sec^2(x)$ or $\sec(x)\tan(x)$, and both substitution paths work well. You also demonstrated integration by parts. While that can technically work, I would recommend skipping it here. It is harder for students to generalize and less obvious how to choose parts. In this section, focusing on one broadly applicable technique will likely reduce confusion and build confidence.

4. *Board visibility*

I would suggest writing larger and spreading your work out more (or enlarging content on the screen if using prepared notes). That makes it easier to point to specific parts and ask:

- “What comes next?”
- “What identity are we using?”
- “What should we label here?”

You are somewhat constrained by the room setup, but small adjustments in spacing and size can make interaction easier.

Overall Impression

This was a strong and productive section. Students were working seriously and engaging with the math. Your early circulation and calm presence are clear strengths. The next step be increasing intentional whole-class interaction. You are building a solid classroom presence. Thank you for your thoughtful work with our students.